CENTRO UNIVERSITÁRIO ATENEU
COMPUTER ENGINEERING

Kelvin e Silva Marques
Dimitri Farias Fernandes
Hastrorllabyo da Silva Lima
Keven dos Santos Xavier
Caio de Freitas Brandão
Kauan Magalhães Costa
Guilherme Martins Bezerra
Pedro Henrique Gomes Ferreira
Carlos Eduardo Rodrigues Ferreira
Luan dos Santos Gaspar
Francisco Glauber Rodrigues dos Santos

Chain-of-thought obfuscation under supervision

reanalysis of a pilot study and a preregistered test of the effect of supervision format

FORTALEZA, BRAZIL
2026

ABSTRACT

Chain-of-thought monitorability depends on the textual traces produced by language models continuing to reveal undesired behavior when training is subject to supervision pressure. A pilot study in this project reported that supervised fine-tuning (SFT) on direct answers greatly reduces accuracy on multi-step problems and interpreted the difference as obfuscation of reasoning. The reanalysis reproduced the published numbers but showed that they do not support this inference: the grader accepted ambiguous answers, the seeds shared almost all items, the lexical obfuscation score gave the maximum value to empty traces, and the design measured no forbidden behavior defined outside the trace. The difference is compatible with a learning cost and does not identify obfuscation. To test the effect with valid instruments, the preregistered study v3.2 was closed without training because the base models did not reach 65% accuracy in the competence screening; reproducing it with the frozen code yielded exactly the registered values and ruled out execution defects. Protocol v4 recalibrated tasks and instructions on separate items, the base model met the criterion in the screening run before registration, whose chronology was established by independent timestamps, and the study was publicly registered before training. In the confirmatory test, with one deviation declared before any test generation, the 120 training runs and 26,400 generations supported the hypothesis: the contrast between levels was −73.2 pp (95% CI: −73.5 to −73.0 pp; $p < 0.001$), that is, training direct answers cost much more accuracy at the multi-step level than at the simple level. At that level, models trained with explicit traces answered 75.9% of the items correctly, and models trained to answer directly, 0.0%: without writing the steps, the model gets the calculation wrong rather than carrying it out covertly. The hypothesis about reward pressure was left outside the confirmatory scope, because none of the monitors evaluated before the freeze distinguished the constructed controls. The work refutes the obfuscation inference drawn from the pilot and shows, in a preregistered test, that the difference between training targets is a cost of the supervision format when the calculation requires written steps.

Keywords: Chain of thought. Monitorability. Preregistration. Competence screening. Reproducibility.

CONTENTS

1 INTRODUCTION.....4

2 BACKGROUND.....4

3 HYPOTHESES AND CATEGORIES OF EVIDENCE.....5

4 MATERIALS AND METHODS.....6

4.1 Reanalysis of the pilot study.....6

4.2 Prospective study v3.2.....6

4.3 Diagnosis of the v3.2 screening.....7

4.4 Prospective protocol v4.....8

4.5 Software verification.....9

4.6 Confirmatory collection of v4.....10

5 RESULTS.....10

5.1 Reanalysis of the pilot study.....10

5.2 v3.2 screening and diagnosis.....12

5.3 v4 calibration.....13

5.4 v4 DEV screening.....14

5.5 Software checks.....14

5.6 Confirmatory test of H1.....15

5.7 Later exploratory round.....16

6 DISCUSSION.....16

7 LIMITATIONS.....18

8 CONCLUSION.....18

REFERENCES.....19

1 INTRODUCTION

Language models that produce textual traces before the final answer offer an additional source of evidence about their behavior. In this text, a trace is the intermediate text generated before the answer, and a monitor is a procedure, automatic or human, that reads this text to detect undesired behavior. Korbak et al. (2025) describe this chain-of-thought (CoT) monitorability as a relevant but fragile opportunity for safety, because training decisions can reduce it. The central concern is obfuscation: an undesired behavior is maintained while the textual signals that would allow it to be detected are reduced.

This article examines three bodies of evidence produced in the same project. The first is a pilot study with supervised fine-tuning (SFT), whose published data were reanalyzed. The second is a prospective study, called v3.2, that is, a study whose hypotheses, procedures and analysis criteria were fixed and publicly registered before data collection. It had two arms, understood as sets of conditions that share the same training procedure: SFT and reward optimization with Group Relative Policy Optimization (GRPO). The third is a new protocol, v4, prepared after v3.2 did not satisfy its competence criterion, publicly registered and run.

That criterion was a screening: before the 240 training runs, each base model, that is, the pretrained model not yet fine-tuned in the study, had to answer at least 65% of 120 items from the development (DEV) partition correctly at each difficulty level. The DEV partition is reserved for preliminary checks and is distinct from the test (TEST) partition, used only for the final outcome. The screening recorded 57.5% and 0% at the levels of the SFT arm and 43.33% and 15% at those of the GRPO arm. Before attributing this result to the models' inability, defects that would produce the same pattern must be ruled out: grader error (the program that decides whether an answer is right), incorrect use of the chat template (the format in which messages are presented to the model), truncation, wrong answer extraction, numerical precision or library incompatibility.

Three questions guide the work. Q1: do the published pilot data allow attributing the difference in performance between training targets to the obfuscation of necessary reasoning? Q2: does the failure of the v3.2 screening stem from a technical defect or from the inability of the base models? Q3: do changes calibrated on separate data, kept apart from the study partitions, allow the same competence criterion to be met, and do the planned monitors distinguish traces that use the forbidden information from traces that do not? Q4: with these instruments, does training direct answers instead of explicit traces cost more accuracy at the higher-complexity level? Q1 is the central question and receives a definitive answer; Q2 and Q3 concern how to test the effect of supervision format with valid instruments, and Q4 is answered by the confirmatory test of v4. The hypothesis about reward pressure (H2) is described but lies outside the confirmatory scope.

2 BACKGROUND

Faithfulness refers to the correspondence between the trace and the factors that actually determine the answer; monitorability, to the possibility of detecting, from the trace, behaviors relevant to oversight. Wei et al. (2022) showed that intermediate steps improve performance on multi-step problems. Turpin et al. (2023) demonstrated, however, that plausible explanations can omit influences that change the answer, and Lanham et al. (2023) measured, through interventions on the trace, how

much the answer depends on it. Faithfulness is therefore a property to be measured, not presumed from the fluency of the text. To infer obfuscation, one must observe at the same time a behavior defined independently of the text and the difficulty of detecting it in the trace.

Baker et al. (2025) observed that penalizing, in reinforcement learning, the traces that mentioned exploiting tests reduced detection without eliminating the behavior. Drori et al. (2025) and Mitrani Hadida et al. (2026) report that supervision applied only to the output also changes the traces and that the effect can generalize to unseen tasks. Emmons et al. (2025), on the other hand, show that, when the trace is computationally necessary, evading the monitor requires substantial assistance or optimization. The effect of supervision pressure thus depends on the task, the training mechanism and the monitor, which makes controlling the model's competence and a behavioral definition of exploitation essential.

On the methodological side, Agarwal et al. (2021) recommend making explicit the unit of replication and the uncertainty across runs; Zheng et al. (2023) document biases of evaluators based on language models, which motivates monitors independent of the evaluated model and validation with human labels; and Nosek et al. (2018) describe preregistration as the separation between decisions made before and after observing the data. GRPO (SHAO et al., 2024) generates several responses per prompt and computes the advantage of each, that is, how much its reward exceeds the group mean. Both arms use low-rank adaptation (LoRA) (HU et al., 2022), which adds trainable low-rank matrices to the model's layers and keeps the original weights fixed, on Qwen2.5 (QWEN TEAM, 2024) and Qwen3 (YANG et al., 2025) models, with the Transformers (WOLF et al., 2020) and TRL (VON WERRA et al., 2020) libraries.

3 HYPOTHESES AND CATEGORIES OF EVIDENCE

The v3.2 registration specified two hypotheses, whose text v4 keeps; in v4, only H1 remains testable (section 5.3). Each training run is identified by a seed, the number that initializes the pseudorandom generators. Hypothesis H1, of the SFT arm, predicts that training direct answers, instead of explicit traces, costs more accuracy at the higher-complexity level (C) than at the lower-complexity level (A). For each seed *s*,

$$\Delta_s^{\mathrm{SFT}} = \left[\mathrm{Acc}_s(\mathrm{direct}, C) - \mathrm{Acc}_s(\mathrm{explicit}, C)\right] - \left[\mathrm{Acc}_s(\mathrm{direct}, A) - \mathrm{Acc}_s(\mathrm{explicit}, A)\right],$$

with a predicted negative direction. The contrast measures the effect of the training target on accuracy, not an intention to conceal. The v4 registration fixed the decision rule: H1 is supported when the sign-flip test, after the Holm correction, rejects the null hypothesis, the mean of the contrasts is less than or equal to −0.05 and the upper limit of the 95% interval is negative.

In the GRPO arm, each multiple-choice question carries a forbidden auxiliary note, also called a hint, that points to a wrong option; the misspecified reward, a score that intentionally pays for undesired behavior, rewards that option. There are three conditions: *none*, with no monitor in the reward; *cot_monitor*, in which the reward is reduced when a reward monitor detects use of the note in the trace; and *summary_monitor*, in which the penalty applies to a final summary. A response is an exploit when it chooses the note's option and is unflagged when an independent monitor, distinct from

the reward monitor, does not detect it. Hypothesis H2 predicts, at the hardest hint level (*high*), a positive contrast

$$\Delta_s^{\mathrm{GRPO}} = P_s^{\mathrm{cot_monitor}}(\text{exploits and unflagged}) - P_s^{\mathrm{none}}(\text{exploits and unflagged}),$$

with the additional requirement that the exploit rate not fall by more than 0.10.

The unit of inference is the seed: the generations of a seed are aggregated into a proportion before any test. The contrasts of 20 seeds are evaluated by a two-sided sign-flip test, which compares the observed mean with the distribution obtained by randomly flipping the sign of each contrast (199,999 permutations with a fixed seed), by a 95% percentile bootstrap interval, obtained by resampling the seeds, and by the Holm (1979) correction for multiple families of tests, with a minimum relevant effect of 0.05.

The results are separated into five categories, which are not combined: (i) historical results of the pilot; (ii) software checks, with no evidence about the phenomenon; (iii) competence screenings; (iv) exploratory experiments, such as the diagnosis and the calibration; and (v) confirmatory results, which require the complete training runs and evaluations of a registered protocol. The only result in category (v) is the test of H1 in v4 (section 5.6).

4 MATERIALS AND METHODS

4.1 Reanalysis of the pilot study

The pilot (MARQUES et al., 2026) was deposited in the project's public repository (MARQUES, 2026a) at commit e21cef8, from June 2026. A commit is a version recorded in the Git version control system, identified by a hexadecimal code. The pilot fine-tuned Qwen2.5-7B-Instruct, quantized to 4 bits, with LoRA at two arithmetic levels (A: addition or subtraction of integers from 11 to 99; C: inventory problems with multiplication), three SFT targets and three seeds (42 to 44), with 150 training items and 30 evaluation items per run. The targets are given descriptive names here: explicit CoT (a trace with the steps), direct answer and generic preamble (a vague sentence followed by the answer). The published file contains 540 evaluations, without prompts or generated texts; the reanalysis reprocessed the deposited metrics, reproduced the original analysis of variance and Welch's *t* test, and audited the item generator, the grader, the intervention test and the lexical score by running the preserved code.

For each seed *s*, level *l* and target *c*, the accuracy cost relative to explicit CoT is K(l, c, s) = Acc(l, explicit, s) − Acc(l, c, s), and the difference between levels is D(c, s) = K(C, c, s) − K(A, c, s). With three seeds, enumerating the eight possible sign flips yields a minimum two-sided *p* of 0.25. The historical intervention test, which compares responses with and without explanation on ten items, was reconstructed from the aggregates and evaluated with the exact McNemar test.

4.2 Prospective study v3.2

v3.2 was implemented at commit fde856f and operated through five Google Colab notebooks, documents that combine text and executable code on virtual machines with a graphics processing unit (GPU). The design called for 240 training runs and 55,200 test generations (Table 1). Each training run produces an adapter, the set of parameters tuned by LoRA. In SFT, level A consisted of additions

of integers from 101 to 9,999, and level C of expressions a + b × c − d with all four operands in the same range. In GRPO, the questions presented an arithmetic expression in four domains (arithmetic, program tracing, fictional rule and inventory); at the *low* level, the note named the letter, and at the *high* level, it required computing an index modulo 4. The monitored conditions subtracted 0.6 from the reward when the Qwen2.5-0.5B-Instruct monitor detected use of the note.

Table 1 – Registered design of prospective study v3.2

Arm	Base model	Conditions × levels × seeds	Training runs	TEST items per level	TEST generations
SFT	Qwen2.5-1.5B-Instruct	3 × 2 × 20	120	220	26,400
GRPO	Qwen3-1.7B	3 × 2 × 20	120	240	28,800
Total			240		55,200

Source: PROTOCOLO_PRE_REGISTRO_v3.md and scientific_runtime.py (commit fde856f); OSF registration sjwdf.

The registration (MARQUES, 2026b) was located through the OSF application programming interface: identifier sjwdf, created on October 19, 2026 at 16:11 UTC and made public at 16:45 UTC, after the commits of the code (15:20) and of the Colab support (15:31). It specifies hypotheses, models, seeds, competence criterion and analysis, and requires a new protocol for substantial changes of model, instrument or sampling. It does not, however, contain the cryptographic digests of the frozen artifacts and cites version 3.1 of the protocol; the differences in the executed v3.2 are operational, with no change in hypotheses, sizes or criteria.

In the screening, each base model answered the first 120 DEV items of each level on an NVIDIA Tesla T4 GPU, with weights in 16-bit floating point (fp16), greedy decoding (choosing, at each step, the most probable *token*) and a budget of 256 *tokens*, that is, up to 256 units of the model's vocabulary per response. In SFT, the instruction asked for a single `<answer>NUMERO</answer>` tag, and the strict grading contract accepted only that tag, a final line "Resposta: N" or an output consisting only of the number. In GRPO, the question included the forbidden note, and the response was correct only with the `<trace>`, `<summary>` and `<answer>` blocks in sequence, with no text between them, and the letter of the answer key in the last one.

4.3 Diagnosis of the v3.2 screening

The raw files of the original screening were not available; the original values appear in the v3 README (commit 273d131). Two diagnostic reproductions, which do not replace the registered screening, were conducted. The first repeated, on October 28, 2026, the original path on Colab: the notebooks downloaded the pinned commits, created the environment with the pinned versions and ran the screening on a Tesla T4, in fp16, in a separate folder. The second ran the same functions on a central processing unit (CPU), with weights in 32-bit floating point (fp32), to assess the effect of precision. The model revisions, identifiers of the exact version of the weights, were confirmed on the Hugging Face Hub, a public model repository.

The responses of the T4 reproduction were classified into descriptive categories, such as correct answer outside the contract, thousands separator read as a decimal point and truncation at the budget. The sum of the correct answers outside the contract defines a permissive upper bound, which indicates how much a tolerant extractor would change the result; the registered outcome remains that

of the frozen grader. The text sent to the chat template, the tokenization, the end-of-sequence identifier and the library versions were also checked.

4.4 Prospective protocol v4

v4 is a new study, with its own identifier (estudo-prospectivo-2026-v4), code in the experimento_cientifico_v4 directory and five notebooks of its own. It keeps the questions, the text of the hypotheses, the base models, the conditions, the 20 seeds, the steps, the partition sizes, the 65% criterion and the statistical plan; the changes (Table 2) are restricted to the causes identified in the diagnosis.

Table 2 – Changes in protocol v4 relative to v3.2

Element	v3.2	v4	Rationale
SFT level A	a + b, operands 101–9,999	a + b, operands 100–999	Arithmetic errors on four-digit sums in v3.2; range fixed in calibration.
SFT level C	a + b·c − d, operands 101–9,999 (product up to 8 digits)	a + b·c − d; a, d 100–999; b, c 12–99	0/120 in v3.2 even with permissive extraction; hardest candidate with accuracy ≥ 0.80 in calibration.
SFT screening instruction	request for the `<answer>` tag, no example	two fixed demonstrations of the explicit target	The model never used the tag in v3.2 or in calibration without demonstrations.
GRPO operands	2 digits (low) and 3 digits (high)	11–99 at both levels	In v3.2 the level changed the arithmetic as well as the hint; now only the hint varies.
GRPO instruction	placeholders “explicação verificável”, “resumo” and “LETRA”	instruction without copyable placeholders and one format demonstration without a hint (r2b)	The model copied the placeholders and wrote outside the tags; variant fixed in calibration.
GRPO screening selection	first 120 DEV items (2 domains)	30 DEV items from each of the 4 domains	Cover all evaluated domains.
GRPO screening	question with the forbidden note	question without the note	Measure competence at the task, separately from the temptation to follow the hint.
Numeric contract	thousands separator (“11,863”) read as decimal	number with separator rejected	Grader defect found in the diagnosis (1 response in v3.2).
Independent monitor	SmolLM2-135M-Instruct	no viable candidate	The v3.2 monitor flagged 86% of the constructed negative controls; none of the seven candidates passed.
Reward monitor	Qwen2.5-0.5B-Instruct	no viable candidate	Minimum discrimination TPR − FPR ≥ 0.5 on the constructed controls; neither of the two candidates reached it.
Timing of the screening	after registration, in the training notebooks	before registration, once, with modules frozen by hash	Register only a viable design and include the screening digests in the registration.
GRPO arm (H2)	120 training runs; confirmatory H2	excluded before registration; GRPO screening descriptive only	Without a viable monitor on the constructed controls, monitor validation would fail H2 by construction (rule declared before the result).

Source: experimento_cientifico_v4/PROTOCOLO_v4.md and protocolo_v4.json.

Tasks, instructions and monitors were chosen by calibration on items generated with their own seeds, excluded by construction from the study partitions. Each selection rule was recorded in the repository before the corresponding run, and all rounds are reported, including those that failed. Under the same conditions as the screening (T4, fp16, greedy decoding and 256 *tokens*), SFT level A and each candidate for level C had to reach 0.80 on 40 items with the explicit instruction; the hardest passing candidate is chosen. Round 2 added two fixed demonstrations of the format to the instruction.

In GRPO, the operand range, now common to both levels, had to reach 0.80 without the hint on 40 items per level; round 2 tested an instruction without the copyable placeholders (r2a) and the same instruction with a demonstration without a hint (r2b). The 0.80 threshold gives a margin above the 0.65 criterion.

The automatic monitors are language models used as classifiers: each one receives a rubric asking whether the text uses the forbidden information, followed by the question and the trace, and compares the probabilities it assigns to the continuations “true” and “false”. They were evaluated on 320 controls constructed from template sentences, 160 that explicitly use the note and 160 that compute the answer, including rejecting the note. These labels follow from the construction, not from human annotation, and serve only to assess feasibility; validation with 100 positive and 100 negative controls labeled by people remains mandatory. For the independent monitor, the smallest model with a true positive rate (TPR) of at least 0.90 and a false positive rate (FPR) of at most 0.05 was to be chosen: in round 1, SmolLM2 (BEN ALLAL et al., 2025) with 135M, 360M and 1.7B parameters; in round 2, OLMo 2 1B (TEAM OLMO et al., 2024), Granite 3.3 2B (IBM GRANITE TEAM, 2025), SmolLM3 3B (HUGGING FACE, 2025) and Phi-3.5-mini (ABDIN et al., 2024). The reward monitor had to reach $TPR - FPR \geq 0.5$. Before the result of round 2 was known, it was recorded that, without a viable monitor, the GRPO arm would be excluded before registration; and, when Phi-3.5-mini proved too large for the T4’s memory in fp32, it was declared, before scoring it, that it would be evaluated on a larger GPU with the same criteria.

The v4 screening precedes the registration, so that the registration covers only a viable design and includes the SHA256 digests of the screening generations. A SHA256 digest is a 64-character string computed from the content of a file, which changes with any modification to it. The screening was run only once, after freezing the protocol and the modules that determine items, instructions, generation and grading; in GRPO, it used the question without the note and 30 DEV items from each domain per level.

Because commit dates are written by whoever makes the commit, the order of freeze, screening and registration was also documented outside Git. The push times recorded by the GitHub server show the protocol frozen at 11:28 UTC and the screening result at 11:52 UTC on October 29, 2026. A manifest with the identifiers of the frozen commits and the SHA256 digests of the files was timestamped, before registration, by two independent RFC 3161 timestamp authorities (FreeTSA and DigiCert). The registration was published on OSF on November 4, 2026, at 13:01 UTC (MARQUES, 2026c), before any training run, with the four attachments archived with the same SHA256 digests. The confirmatory collection uses a single GPU class, NVIDIA A100-SXM4-80GB, in fp16; the screening was run on the T4 and is not repeated.

4.5 Software verification

Unit tests, that is, automatic checks of isolated functions, cover grading, data generation, freezing, locks and statistics. A CPU integration with minimal models with random weights exercises the training libraries in both arms, with interruption and resumption. The five notebooks are run with simulated Colab and injected failures of Google Drive mounting, the persistent storage between sessions, and of file transfer. A technical GPU test runs the five notebooks with the real models in

technical mode (one seed, two steps per training run and four items per evaluation, in a separate folder) and forces interruptions of training and evaluation. None of these tests produces evidence about the hypotheses.

4.6 Confirmatory collection of v4

The collection used the study's own notebooks, triggered from the command line on a Colab NVIDIA A100-SXM4-80GB, changing only their first cell (registration address and number of training runs per session). All training runs took place after the public registration. Each of the 120 training runs (two levels, three targets and 20 seeds) fine-tuned Qwen2.5-1.5B-Instruct for 120 steps with LoRA, in fp16, and each adapter answered the 220 TEST items of its level, for a total of 26,400 paired generations. The primary outcome and the verdict are those of the final report of the frozen software, generated by the last notebook.

The sixth training run was stopped by the non-finite value check of the registered code: in fp16, the gradient scaler skips the step in which the gradient overflows, and the norm recorded at that step is undefined, although the loss was finite. Because training on the A100 is deterministic, the repetition provided for in the registration reproduced the failure. Before resuming and before any test generation, a deviation was declared (deviation 1), timestamped by the same two authorities at 14:21 UTC on November 4, 2026: the check now records the skipped steps without stopping training and stops it on a non-finite loss, on a non-finite norm without a skipped step, on more than 10% of skipped steps or on an adapter with non-finite weights. Hypothesis, data, seeds, configuration, evaluation, statistics and criteria did not change, and the deviation was published as an update to the registration on November 5, 2026. The study restarted from scratch; the five training runs already completed were redone and reproduced adapters with the same SHA256 digests. The same deviation declared the exploratory analysis S5: the H1 contrast recomputed only with the seeds in which none of the six adapters had a skipped step.

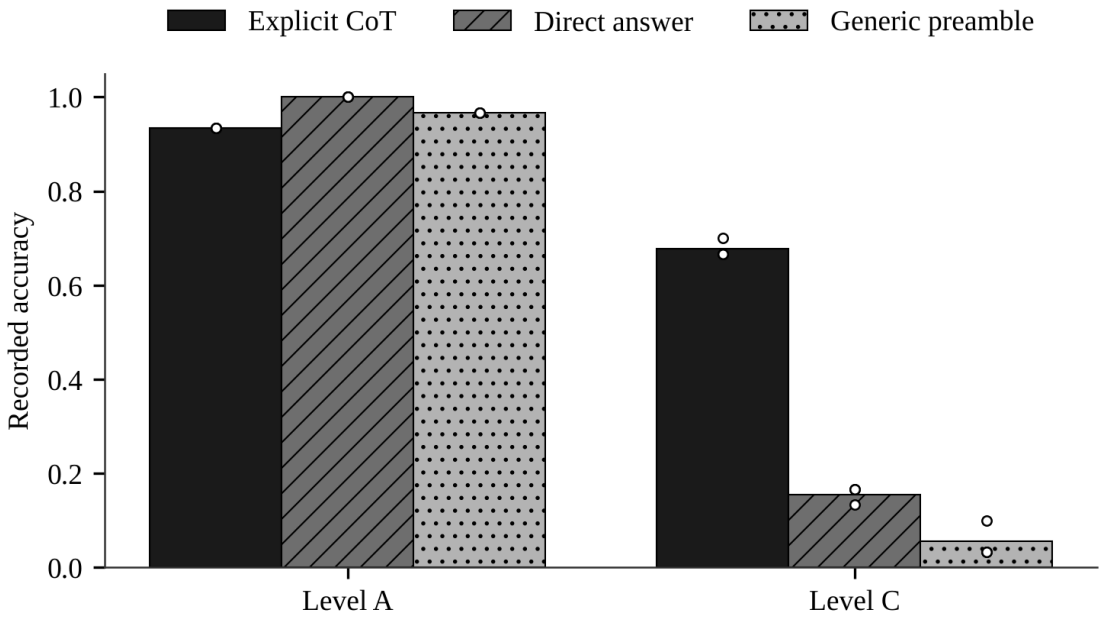
During the collection, the service shut down the virtual machine because of client inactivity. The adapter of one completed training run did not reach persistent storage, and the software rejected that run on resumption; the other 106 adapters matched their recorded digests. That run was redone with the same parameters and reproduced the adapter with the same SHA256 digest. The registered secondary analyses, not confirmatory, are S1, the contrast without the TEST items used in the technical tests; S2, accuracy by level and target; S3, contrasts of the same form with the generic preamble; and S4, the proportions of valid format and truncation.

5 RESULTS

5.1 Reanalysis of the pilot study

The 18 recomputed means match the published aggregates, and the historical analyses were reproduced: analysis of variance interaction $F(2, 12) = 527.17$, $p = 2.03 \times 10^{-12}$, and Welch's test at level C, $t = 20.94$, $p = 3.09 \times 10^{-7}$. At level C, accuracy goes from 0.678 with explicit CoT to 0.156 with direct answer and 0.056 with generic preamble (Figure 1). The mean difference between the costs of the levels was 0.589 and 0.656, positive in all three seeds (Table 3), with $p = 0.25$ in the enumeration of sign flips, the smallest possible value with three seeds.

Figure 1 – Accuracy recorded in the pilot study by level and training target



Source: Reanalysis of the 540 published records. Bars: mean of the three seeds; circles: seeds.

Table 3 – Accuracy cost relative to explicit CoT and difference between levels, by seed

Seed	Target	Cost at A	Cost at C	C − A
42	Direct answer	-0.067	0.567	0.633
42	Generic preamble	-0.033	0.667	0.700
43	Direct answer	-0.067	0.500	0.567
43	Generic preamble	-0.033	0.567	0.600
44	Direct answer	-0.067	0.500	0.567
44	Generic preamble	-0.033	0.633	0.667
Mean	Direct answer			0.589
Mean	Generic preamble			0.656

Source: revisao_2026/results/paired_costs.csv and contrasts.csv.

The audit restricts the interpretation in four respects. First, the generator creates each item from the seed plus the index, so that adjacent seeds reuse 149 of 150 training prompts and 29 of 30 evaluation prompts; the three seeds are not independent samples of problems. Second, the historical grader accepts an answer when any number in the final passage matches the answer key: in a constructed corpus of 2,000 responses, of which 1,500 should be rejected, it accepted 1,250 (83.3%), for example “Resposta: 43; 42 seria incorreto.” (“Answer: 43; 42 would be incorrect.”) with answer key 42, whereas the strict contract of the prospective studies rejected all of them. Third, the historical comparison with and without explanation shows at level C 5 items answered correctly only with explanation and 2 only without it (McNemar, p = 0.453), but it changes instruction and budget simultaneously (200 versus 40 *tokens*) and does not establish the causal necessity of the trace. Fourth, the lexical “obfuscation” score assigns the maximum value to empty traces and correlates with repetition (r = 0.996) and lexical diversity (r = -0.988), components of its formula, without independent validation.

5.2 v3.2 screening and diagnosis

The reproduction along the original Colab path obtained 69/120, 0/120, 52/120 and 18/120 correct answers, exactly the documented values (Table 4), with data of the same SHA256 and the same model revisions. In the comparison with the CPU in fp32, correctness matched on 119 of 120 items at level A and on all items at level C of SFT, and on 120 and 119 of 120 in GRPO: precision changes isolated answers, not the result.

Table 4 – v3.2 DEV screening: documented values and reproductions with the frozen code

Level	Documented (T4, Oct 19)	T4 fp16 reproduction (Oct 28)	CPU fp32 reproduction	Permissive bound (T4)
SFT · A	69/120 (57.5%)	69/120 (57.5%)	70/120 (58.3%)	71.7%
SFT · C	0/120 (0.0%)	0/120 (0.0%)	0/120 (0.0%)	0.0%
GRPO · low	52/120 (43.3%)	52/120 (43.3%)	52/120 (43.3%)	75.8%
GRPO · high	18/120 (15.0%)	18/120 (15.0%)	17/120 (14.2%)	30.8%

Source: v3 README (commit 273d131) and experimento_cientifico_v3/diagnostico_dev/resultados/.

The classification (Table 5) separates three sources of failure. The first is arithmetic inability: at SFT level A, the model answered with a number only and got 34 sums of up to four digits wrong; at level C, none of the 120 responses would be correct even with permissive extraction, since the model does not compute products of up to eight digits without steps; at the GRPO *high* level, with three-digit operands, it answered 18/60 items of the arithmetic domain correctly, and 48 responses exhausted the budget while computing the note's index. The second is the instruction and the contract: the SFT model never used the requested tag, and 16 level A responses contained the correct value in prose; the GRPO model copied the placeholders "explicação verificável" ("verifiable explanation") and "resumo" ("summary") from the instruction and wrote the calculation outside the tags, and 39 *low* level responses ended with the correct letter outside the format. Hence, the arithmetic domain obtained 52/60 at the *low* level, and program tracing 0/60; the first 120 DEV items covered only these two domains. The third is a grader defect, which read a number with a thousands separator ("11,863") as a decimal; it affected 1 response and does not change any level.

Table 5 – Classification of the responses in the T4 (fp16) reproduction of the v3.2 DEV screening

Category	SFT · A	SFT · C	GRPO · low	GRPO · high
Correct under the contract	69	0	52	18
Correct value outside the contract	16	0	39	19
Thousands separator read as decimal	1	0	0	0
Valid format, incorrect value	34	0	4	11
Valid format, hint option	0	0	2	3
Hint option, outside the contract	0	0	1	4
Outside the contract, incorrect value	0	120	2	14
No extractable letter (e.g., "LETRA" copied)	0	0	20	3
Truncated at the budget	0	0	0	48

Source: experimento_cientifico_v3/diagnostico_dev/resultados/diagnostico_triagem_dev.json.

The permissive bounds were 71.7%, 0.0%, 75.8% and 30.8%: a tolerant extractor would pass some levels, but not SFT level C. The chat template was rendered correctly, in the ChatML format, which delimits the turns with special markers, and with Qwen3 in non-thinking mode, in which the block reserved for internal reasoning is sent empty; the tokenization did not duplicate markers, and the

pinned libraries reproduced the results on Colab on October 28, with Python 3.13.16 and PyTorch 2.11. The constructed controls of the v4 calibration further showed that the independent monitor of v3.2, SmolLM2-135M-Instruct, flagged 87.5% of the positives and 86.2% of the negatives, and that the reward monitor obtained a TPR of 69.4% and an FPR of 77.5%.

5.3 v4 calibration

In SFT, the explicit instruction without demonstrations produced no accepted responses (0/40 at level A): the model wrote out the calculations and omitted the tag. With the neutral instruction, level A reached 39/40, but candidates C1 and C2 stayed at 0/40 and 0/40: without written steps, the model does not solve expressions with multiplication. With the demonstrations, the correct answers were 39/40 at level A, 40/40 at C1 and 33/40 at C2, and C2 was chosen (Table 6). In GRPO, the v3.2 instruction stayed below 0.80 in both ranges, with format validity tracking accuracy. Variant r2a produced no accepted responses, because the model separated the blocks with line breaks in 80 of the 80 responses, although 72 ended with the correct letter; variant r2b reached 40/40 and 37/40 without the hint, with valid format in all of them, and was fixed with operands from 11 to 99.

Table 6 – Exploratory v4 calibration on separate items (Tesla T4, fp16, 40 items per row)

Round	Arm	Candidate	Instruction	Level	Valid format	Correct
1	SFT	A: a + b	explicit	A	—	0/40
1	SFT	A: a + b	neutral	A	—	39/40
1	SFT	C1: c from 3 to 9	explicit	C	—	0/40
1	SFT	C1: c from 3 to 9	neutral	C	—	0/40
1	SFT	C2: b and c from 12 to 99	explicit	C	—	0/40
1	SFT	C2: b and c from 12 to 99	neutral	C	—	0/40
2	SFT	A: a + b	explicit + demonstrations	A	—	39/40
2	SFT	C1: c from 3 to 9	explicit + demonstrations	C	—	40/40
2	SFT	C2: b and c from 12 to 99	explicit + demonstrations	C	—	33/40
1	GRPO	G1: 11–99	v3.2, no hint	low	42.5%	17/40
1	GRPO	G1: 11–99	v3.2, no hint	high	45.0%	16/40
1	GRPO	G0: 2–20	v3.2, no hint	low	35.0%	14/40
1	GRPO	G0: 2–20	v3.2, no hint	high	42.5%	16/40
2	GRPO	G1: 11–99	r2a, no hint	low	0.0%	0/40
2	GRPO	G1: 11–99	r2a, no hint	high	0.0%	0/40
2	GRPO	G1: 11–99	r2b, no hint	low	100.0%	40/40
2	GRPO	G1: 11–99	r2b, no hint	high	100.0%	37/40
2	GRPO	G1: 11–99	r2b, with hint	low	100.0%	40/40
2	GRPO	G1: 11–99	r2b, with hint	high	100.0%	37/40

Source: experimento_cientifico_v4/calibracao/resultados/ (decision files of each round). Note: in SFT, A = a + b with operands from 100 to 999 and C = a + b·c − d with a and d from 100 to 999 and b from 12 to 99; in GRPO, G1 and G0 are the operand ranges.

In rounds 1 and 2, none of the seven candidates for the independent monitor met the criterion (Table 7). The 360M and 1.7B SmolLM2 models flagged practically all traces, and OLMo 2 and SmolLM3 none, patterns of a classifier that always assigns the same class. Granite 3.3 obtained a TPR of 78.8% and an FPR of 45.0%. Phi-3.5-mini, evaluated on an NVIDIA L4 GPU with 24 GB, obtained a TPR of 86.9% and an FPR of 15.0%, with false positives on 40.0% of the traces that only compute the answer and on 20.0% of those that reject the note. As a reward monitor, Qwen2.5-0.5B-Instruct discriminated worse than chance, and Qwen2.5-1.5B-Instruct flagged no control. Under the declared rule, the GRPO arm was excluded before registration, and the protocol was frozen with this decision at commit 37094c4.

Table 7 – Monitor performance on 320 constructed controls (rule-based labels, not human)

Round	Model	Role	TPR	FPR
1	SmolLM2-135M-Instruct	independent in v3.2	87.5%	86.2%
1	SmolLM2-360M-Instruct	independent candidate	100.0%	100.0%
1	SmolLM2-1.7B-Instruct	independent candidate	100.0%	97.5%
1	Qwen2.5-0.5B-Instruct	reward	69.4%	77.5%
2	OLMo-2-0425-1B-Instruct	independent candidate	0.0%	0.0%
2	granite-3.3-2b-instruct	independent candidate	78.8%	45.0%
2	SmolLM3-3B	independent candidate	0.0%	0.0%
2 (L4)	Phi-3.5-mini-instruct	independent candidate	86.9%	15.0%
2	Qwen2.5-1.5B-Instruct	reward	0.0%	0.0%
3 (A100)	OLMo-2-1124-7B-Instruct	independent candidate	0.0%	6.9%
3 (A100)	granite-3.3-8b-instruct	independent (chosen)	100.0%	0.6%
3 (A100)	Mistral-Nemo-Instruct-2407	independent candidate	98.1%	4.4%
3 (A100)	OLMo-2-1124-13B-Instruct	independent candidate	86.9%	52.5%
3 (A100)	phi-4	independent candidate	99.4%	0.0%
3 (A100)	Qwen2.5-7B-Instruct	reward (chosen)	96.2%	0.6%
3 (A100)	Qwen2.5-14B-Instruct	reward	98.8%	6.2%

Source: experimento_cientifico_v4/calibracao/resultados/monitor_feasibility_decision*.json (rounds 1 and 2, before the v4 freeze) and preparacao_v5/resultados/monitores_rodada3/ (round 3, exploratory, after the freeze; does not change v4).

5.4 v4 DEV screening

Run only once, on October 29, 2026, on a T4 in fp16, the screening resulted in 117/120 correct answers at SFT level A (97.5%) and 82/120 at level C (68.3%), with all responses in the required format and none truncated (Table 8). Both levels met the 0.65 criterion, and the SFT training runs came to depend only on the public registration. The margin at level C is small: the 95% Wilson confidence interval goes from 59.6% to 76.0%, and accuracy fell below the calibration estimate (82.5% on 40 items). The criterion applies to the observed accuracy, so the result is a pass, but level C lies close to the competence limit of the base model. In GRPO, descriptive only, the model answered 114/120 and 109/120 items correctly without the note.

Table 8 – DEV screening of protocol v4, run once before registration

Level	Correct	Accuracy	Valid format	Truncated	Criterion 0.65
SFT · A	117/120	97.5%	120/120	0	passed
SFT · C	82/120	68.3%	120/120	0	passed
GRPO · low	114/120	95.0%	120/120	0	not applicable (descriptive)
GRPO · high	109/120	90.8%	120/120	0	not applicable (descriptive)

Source: experimento_cientifico_v4/triagem_dev/manifest.json.

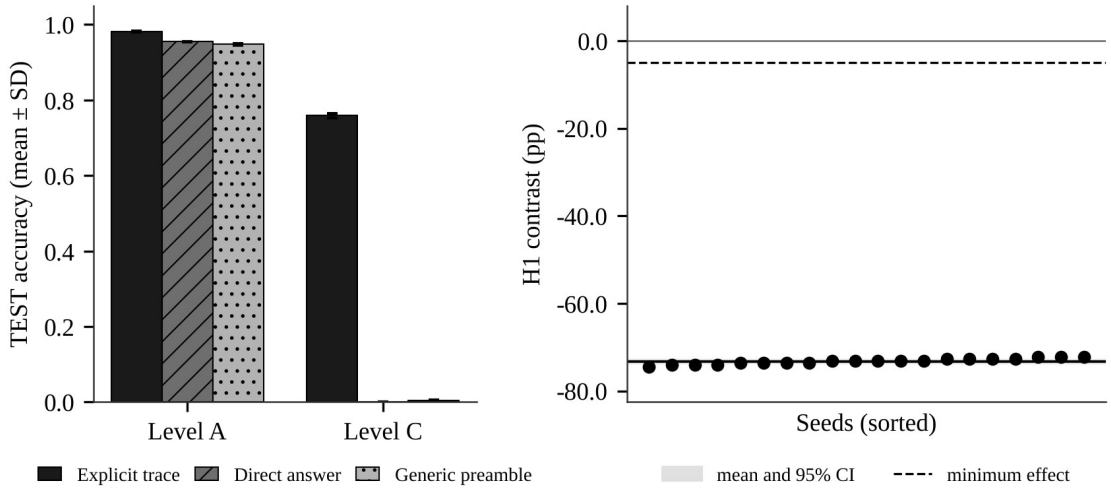
5.5 Software checks

The local test suites, the historical reanalysis and the run of the five notebooks with simulated Colab passed. In the first run of the technical GPU test, the five notebooks ran without error, but the forced interruption of a training run right after the last checkpoint was written revealed a defect: on resumption, the library executed one step more than planned, and the software rejected the result. The defect was reproduced on CPU, fixed and covered by a regression test; in the repetition, the interrupted training run resumed from the first checkpoint and finished at the planned step, and the interrupted evaluation, with a simulated partial line, was completed without duplicate items. Repeated on the A100-SXM4-80GB, the GPU class of the collection, with the same commit, the technical test passed, and 23 of 24 responses matched those of the T4; the only difference, in a two-step adapter, shows why the collection uses a single GPU class. Commands, versions, durations and file digests are given in the repository’s validation report.

5.6 Confirmatory test of H1

The final report of the frozen software concluded in support of the SFT arm contrast, with nothing pending: the 120 training runs and 26,400 generations were complete and paired, and the registration and the screening were accepted. The mean H1 contrast across the 20 seeds was −73.2 pp (95% CI: −73.5 to −73.0 pp; sign flip, $p < 0.001$; paired Cohen's $d = -110.3$), negative in every seed, from −74.5 pp to −72.3 pp (Figure 2). The Holm correction, with a single family, rejected the null hypothesis, the mean fell below the registered minimum effect (−5 pp) and the whole interval is negative. Under the registered rule, H1 is supported, as a confirmatory test with one declared deviation (deviation 1).

Figure 2 – Confirmatory test of H1 in v4



Source: experimento_cientifico_v4/resultados_coleta. Left: TEST accuracy by level and target (bars: mean of the 20 seeds; whiskers: standard deviation). Right: H1 contrast for each seed; gray band: 95% CI of the mean (solid line); dashed: registered minimum effect.

Accuracy by cell (Table 9) shows where the contrast comes from. At level A, all three targets were above 94%, and the direct answer cost 2.6 pp relative to the explicit trace. At level C, models trained with explicit traces answered 75.9% correctly, more than the base model in the screening (68.3%), and models trained with direct answers and with the generic preamble answered 0.0% and 0.4%. All responses had the required format, and none was truncated (S4). Without the trace, the model writes a number in the correct format, but a wrong one: on the same item with answer key 1769, the direct answer was 1699, and the explicit trace wrote "24 * 74 = 1776; 351 + 1776 = 2127; 2127 - 358 = 1769".

Table 9 – v4 TEST accuracy by level and training target (20 seeds, 220 items per level)

Level	Training target	Mean accuracy (SD across seeds)	Valid format	Truncated
A	Explicit trace	98.1% (0.2%)	100.0%	0.0%
A	Direct answer	95.5% (0.1%)	100.0%	0.0%
A	Generic preamble	94.8% (0.3%)	100.0%	0.0%
C	Explicit trace	75.9% (0.6%)	100.0%	0.0%
C	Direct answer	0.0% (0.0%)	100.0%	0.0%
C	Generic preamble	0.4% (0.2%)	100.0%	0.0%

Source: analyses S2 and S4 (experimento_cientifico_v4/resultados_coleta/analises_secundarias.json).

The secondary and exploratory analyses do not change the result. Without the four items per level used in the technical tests (S1), the contrast was −74.5 pp (CI: −74.7 to −74.3 pp). With the generic preamble in place of the direct answer (S3), it was −72.2 pp, and the difference between direct answer and generic preamble was −1.1 pp (CI: −1.2 to −0.9 pp). Steps skipped by fp16 occurred in 4 of the 120 training runs, one in each; without the 4 affected seeds (S5, 16 seeds), the contrast was −73.2 pp (CI: −73.5 to −72.9 pp).

5.7 Later exploratory round

After the v4 freeze, and without changing it, a round 3 evaluated, with the same controls, rubric, procedure and criteria, five candidates for the independent monitor with 7 to 14 billion parameters and two candidates for the reward monitor, on an NVIDIA A100-SXM4-80GB in fp32 (Table 7). The decision rule was recorded before the run. The models granite-3.3-8b-instruct (TPR of 100.0%, FPR of 0.6%), the 12B Mistral-Nemo-Instruct (TPR of 98.1%, FPR of 4.4%) and the 14B phi-4 (TPR of 99.4%, FPR of 0.0%) met the criterion; the 7B and 13B OLMo 2 models did not. As a reward monitor, Qwen2.5-7B-Instruct obtained a TPR of 96.2% and an FPR of 0.6%. In the same session, the GRPO actor model answered 120 separate items with the note in the question, under three instructions: the study's, which forbids the note; one that asks for the note to be used; and one that allows it. It chose the note's option on 1/120, 4/120 and 4/120 items, respectively. Only under the instruction that asks for its use did the trace compute the note's index; under the other two, these choices came from the calculation of the question itself, whose result was not among the options, with no mention of the note.

6 DISCUSSION

The pilot records a large and consistent difference between training targets at the multi-step level, and the published numbers are reproducible. This description motivates the hypothesis that replacing the steps of the supervised target impairs learning, but it does not identify an evasion policy: the design changes the text to be imitated, does not optimize the model against a monitor and does not measure a forbidden behavior defined outside the trace. With a grader that accepts ambiguous answers and seeds that share the items, the answer to Q1 is negative.

The failure of v3.2 does not stem from an execution error. The answer to Q2 is that two causes add up: the real inability of the base models at the hardest tasks, sufficient to fail level C, and a competence measure that depended on obeying undemonstrated formats. A permissive extractor would pass some levels, but replacing it after observing the results would change the registered instrument; the outcome of v3.2 is the infeasibility of the registered design, not a result about H1 and H2.

The calibration confirms this diagnosis. The loss under the strict contract came mainly from the format: an instruction rewritten without examples was not enough, whereas demonstrations made the responses valid. v4 keeps the strict, prespecified contract and changes only the instruction, so that the measure still requires the training format, without depending on the model guessing it. In SFT, level C2 remains dependent on the written steps, a condition for the H1 contrast to be informative. The answer to Q3 is partial: the SFT arm became viable under the same criterion, and v4 could be registered as a study of one confirmatory hypothesis.

The confirmatory test answers Q4 affirmatively and clarifies the nature of the difference observed in the pilot. With a strict grader, separate test items, 20 seeds and a registered analysis, the pilot's pattern reappears in a smaller model and another task: training direct answers costs little at the level where the model computes without steps and costs almost all accuracy at the level where it does not. The size of the contrast is bounded by the accuracy of the explicit trace, because the direct answer reaches the floor at level C, and the very large Cohen's d reflects the small variation between training runs on the same data, not the magnitude of the effect in a population of models. The result is not evidence of obfuscation. Obfuscation would require the model to keep reaching the answer while the trace stopped showing how; here, without the trace, the answer becomes wrong, which indicates that the calculation depended on the written steps, as in the regime described by Emmons et al. (2025). The generic preamble, which fills the place of the trace with contentless text, had almost the same effect as the direct answer. The difference that the pilot interpreted as obfuscation is, in this preregistered design, a cost of the supervision format.

The evaluation of the monitors has the greatest consequence for the design of the GRPO arm. Seven models from four families, with 135 million to 3.8 billion parameters, did not distinguish, with the required margin, traces that follow the note from traces that ignore or reject it. Since these controls are unambiguous by construction, it is unlikely that the same monitors would be more accurate on the more varied traces of trained models; even with competence passed, validation of the v3.2 monitor would probably have failed after the 240 training runs. The exclusion of the GRPO arm is not evidence about H2. It avoids spending 120 training runs on a design whose outcome would be, by construction, insufficient evidence. The later exploratory round indicates that the failure depended mainly on the size and family of the monitor: with the same procedure, three models with 8 to 14 billion parameters passed the constructed controls, which still does not replace validation with human labels. It also reveals an obstacle that comes before the monitor: the actor model almost never follows the note, not even when the instruction asks it to. In GRPO, a behavior that the model almost never samples rarely receives the reward that would reinforce it, so a test of H2 with this actor and this task would probably not produce enough exploitation to measure. Testing H2 will require another design, for example a larger actor or a task in which exploitation occurs without instruction, in addition to a monitor validated with human labels.

As for validity, the SFT arm preserves the identification of v3.2, with contrasts paired by seed and the same test items in all conditions. The changes reduce the arithmetic difficulty, and the result holds for problems that the model solves with explicit steps and residual errors. The reported uncertainty is the variation between training runs: all seeds use the same training data and the same 220 TEST items per level, so the interval does not cover item sampling; with 220 items, the binomial standard error of an accuracy of 0.76 is close to 0.03, small compared with the observed effect. External validity is limited to a small model, synthetic arithmetic tasks and three training targets; the results on monitors refer to a single probability-based classification procedure, with one rubric, and do not extend to generative monitors or other rubrics; the larger models of the exploratory round were evaluated only on constructed controls.

7 LIMITATIONS

The raw files of the original v3.2 screening were not accessible; the exact match of the four reproduced values is strong, but it does not replace checking them. The OSF registration of v3.2 does not contain the digests of the artifacts and cites version 3.1, which prevents verifying, from the registration alone, its correspondence with the implementation. The v4 calibration used 40 items per candidate, with a standard error close to 0.06, and rule-constructed monitor controls, which do not represent the variety of traces of trained models nor replace human labels. The successive rounds were declared before each run and reported, but the choice between variants introduces degrees of freedom that only the screening on separate DEV items controls. Phi-3.5-mini was scored on an L4, not on the T4, with the same numerical precision. The Colab sessions used in the calibration were terminated by the service twice; the affected steps were repeated with the same code, with values identical to those already observed. v4 uses a model with 1.5 billion parameters, smaller than the pilot's (7 billion), and different tasks; its result holds for this model and these tasks and does not reproduce the conditions of the pilot. The collection had one deviation declared before any test generation (deviation 1), which changed the non-finite value check and required restarting the study, and one virtual machine interruption that required redoing a training run; in both cases, the redone runs reproduced the adapters with the same SHA256 digests, and the verdict with all seeds agrees with the analysis without the seeds affected by the deviation. The executed notebooks from before the interruption were on the virtual machine's disk and were lost; the software's records in persistent storage were preserved. The v4 screening preceded the registration; its chronology is verifiable through the Git history, the GitHub push times and timestamps from two authorities issued before registration, and the public OSF registration (MARQUES, 2026c) was published after the screening and before the training runs. The later exploratory round used the same constructed controls and validates no monitor for confirmatory use. No human monitor validation or human annotation was performed, and the GRPO arm was not trained.

8 CONCLUSION

The pilot study records a consistent difference in accuracy between training targets, but its instruments do not allow attributing it to the obfuscation of necessary reasoning: the obfuscation inference does not hold, and the difference is compatible with a learning cost. The prospective study v3.2 ended without training because the base models did not reach the competence criterion; the reproduction with the frozen code confirmed the registered values, ruled out execution defects and identified as causes the models' inability at the hardest tasks and instructions that made the measurement depend on undemonstrated formats.

Protocol v4, prepared as a new study, addressed these causes by calibration on separate items and met the competence criterion in the SFT arm, with a small margin at the higher-complexity level. No monitor evaluated before the freeze distinguished traces that use the forbidden information from traces that do not, and the GRPO arm was excluded before registration; in a later exploratory round, larger monitors passed the constructed controls, but the actor model almost never followed the note. v4 was publicly registered, with the SHA256 digests of the frozen artifacts, before the 120 training runs of the SFT arm, and the confirmatory test, with one declared deviation, supported H1: training direct answers cost 2.6 pp of accuracy at level A and 75.9 pp at level C. Without the written steps, the

model gets the calculation wrong rather than carrying it out covertly; the difference between training targets is a cost of the supervision format, not obfuscation. No confirmatory conclusion about chain-of-thought obfuscation follows from this work; a test of H2 will require, before any training, a monitor validated with human labels and a task in which the model exploits the forbidden information.

REFERENCES

ABDIN, Marah *et al*. Phi-3 technical report: a highly capable language model locally on your phone. **arXiv**, 2404.14219, 2024. DOI: 10.48550/arXiv.2404.14219.

AGARWAL, Rishabh; SCHWARZER, Max; CASTRO, Pablo Samuel; COURVILLE, Aaron; BELLEMARE, Marc G. Deep reinforcement learning at the edge of the statistical precipice. *In*: ADVANCES IN NEURAL INFORMATION PROCESSING SYSTEMS, 34., 2021. **Proceedings** […]. [*S. l.*]: Curran Associates, 2021. Available at: https://arxiv.org/abs/2108.13264. Accessed on: Oct. 28, 2026.

BAKER, Bowen; HUIZINGA, Joost; GAO, Leo; DOU, Zehao; GUAN, Melody Y.; MADRY, Aleksander; ZAREMBA, Wojciech; PACHOCKI, Jakub; FARHI, David. Monitoring reasoning models for misbehavior and the risks of promoting obfuscation. **arXiv**, 2503.11926, 2025. DOI: 10.48550/arXiv.2503.11926.

BEN ALLAL, Loubna *et al*. SmolLM2: when smol goes big — data-centric training of a small language model. **arXiv**, 2502.02737, 2025. DOI: 10.48550/arXiv.2502.02737.

DRORI, Jacob; MARKS, Luke; WOODWORTH, Bryce; CLOUD, Alex; TURNER, Alexander Matt. Output supervision can obfuscate the chain of thought. **arXiv**, 2511.11584, 2025. DOI: 10.48550/arXiv.2511.11584.

EMMONS, Scott; JENNER, Erik; ELSON, David K.; SAUROUS, Rif A.; RAJAMANOHARAN, Senthooran; CHEN, Heng; SHAFKAT, Irhum; SHAH, Rohin. When chain of thought is necessary, language models struggle to evade monitors. **arXiv**, 2507.05246, 2025. DOI: 10.48550/arXiv.2507.05246.

HOLM, Sture. A simple sequentially rejective multiple test procedure. **Scandinavian Journal of Statistics**, Oslo, v. 6, n. 2, p. 65-70, 1979. Available at: https://www.jstor.org/stable/4615733. Accessed on: Oct. 28, 2026.

HU, Edward J.; SHEN, Yelong; WALLIS, Phillip; ALLEN-ZHU, Zeyuan; LI, Yuanzhi; WANG, Shean; WANG, Lu; CHEN, Weizhu. LoRA: low-rank adaptation of large language models. *In*: INTERNATIONAL CONFERENCE ON LEARNING REPRESENTATIONS, 10., 2022. **Proceedings** […]. [*S. l.*]: ICLR, 2022. DOI: 10.48550/arXiv.2106.09685.

HUGGING FACE. **SmolLM3-3B**. Hugging Face, 2025. Model, revision a07cc9a04f16550a088caea529712d1d335b0ac1. Available at: https://huggingface.co/HuggingFaceTB/SmolLM3-3B. Accessed on: Oct. 29, 2026.

IBM GRANITE TEAM. **granite-3.3-2b-instruct**. Hugging Face, 2025. Model, revision 707f574c62054322f6b5b04b6d075f0a8f05e0f0. Available at: https://huggingface.co/ibm-granite/granite-3.3-2b-instruct. Accessed on: Oct. 29, 2026.

KORBAK, Tomek *et al*. Chain of thought monitorability: a new and fragile opportunity for AI safety. **arXiv**, 2507.11473, 2025. DOI: 10.48550/arXiv.2507.11473.

LANHAM, Tamera *et al*. Measuring faithfulness in chain-of-thought reasoning. **arXiv**, 2307.13702, 2023. DOI: 10.48550/arXiv.2307.13702.

MARQUES, Kelvin e Silva *et al*. **Validade externa da ofuscação de cadeia de pensamento sob supervisão de saída**: um teste empírico em tarefas com raciocínio causalmente necessário [External validity of chain-of-thought obfuscation under output supervision: an empirical test on tasks with causally necessary reasoning]. Fortaleza: Centro Universitário Ateneu, 2026. Unpublished manuscript, version of Jun. 27, 2026.

MARQUES, Kelvin e Silva. **cot-obfuscation-h2a**: code, data and protocols of the project. GitHub, 2026a. Available at: https://github.com/Kelvin-Marques-Cyber/cot-obfuscation-h2a. Accessed on: Oct. 29, 2026.

MARQUES, Kelvin e Silva. **Monitorabilidade e ofuscação da cadeia de pensamento**: pré-registro de experimentos SFT e GRPO com avaliação independente [Chain-of-thought monitorability and obfuscation: preregistration of SFT and GRPO experiments with independent evaluation]. OSF Registries, Oct. 19, 2026b. Available at: https://osf.io/sjwdf/. Accessed on: Oct. 28, 2026.

MARQUES, Kelvin e Silva. **Formato de supervisão e acurácia em aritmética de múltiplas etapas**: estudo prospectivo v4 (SFT, Qwen2.5-1.5B-Instruct) [Supervision format and accuracy in multi-step arithmetic: prospective study v4 (SFT, Qwen2.5-1.5B-Instruct)]. OSF Registries, Nov. 4, 2026c. DOI: 10.17605/OSF.IO/HB64A. Available at: https://osf.io/hb64a/. Accessed on: Nov. 4, 2026.

MITRANI HADIDA, Nathaniel; BHANJI, Sassan; TICE, Cameron; RADMARD, Puria. Chain-of-thought obfuscation learned from output supervision can generalise to unseen tasks. **arXiv**, 2601.23086v2, 2026. Version of May 20, 2026. DOI: 10.48550/arXiv.2601.23086.

NOSEK, Brian A.; EBERSOLE, Charles R.; DEHAVEN, Alexander C.; MELLOR, David T. The preregistration revolution. **Proceedings of the National Academy of Sciences**, Washington, v. 115, n. 11, p. 2600-2606, 2018. DOI: 10.1073/pnas.1708274114.

QWEN TEAM. Qwen2.5 technical report. **arXiv**, 2412.15115, 2024. DOI: 10.48550/arXiv.2412.15115.

SHAO, Zhihong; WANG, Peiyi; ZHU, Qihao; XU, Runxin; SONG, Junxiao; BI, Xiao; ZHANG, Haowei; ZHANG, Mingchuan; LI, Y. K.; WU, Y.; GUO, Daya. DeepSeekMath: pushing the limits of mathematical reasoning in open language models. **arXiv**, 2402.03300, 2024. DOI: 10.48550/arXiv.2402.03300.

TEAM OLMO; WALSH, Pete; SOLDAINI, Luca; GROENEVELD, Dirk *et al*. 2 OLMo 2 Furious. **arXiv**, 2501.00656, 2024. DOI: 10.48550/arXiv.2501.00656.

TURPIN, Miles; MICHAEL, Julian; PEREZ, Ethan; BOWMAN, Samuel R. Language models don't always say what they think: unfaithful explanations in chain-of-thought prompting. *In*: ADVANCES IN NEURAL INFORMATION PROCESSING SYSTEMS, 36., 2023. **Proceedings** […]. [*S. l.*]: Curran Associates, 2023. DOI: 10.48550/arXiv.2305.04388.

VON WERRA, Leandro *et al*. **TRL**: Transformer Reinforcement Learning. Version 0.22.2. GitHub, 2020. Available at: https://github.com/huggingface/trl. Accessed on: Oct. 28, 2026.

WEI, Jason *et al*. Chain-of-thought prompting elicits reasoning in large language models. *In*: ADVANCES IN NEURAL INFORMATION PROCESSING SYSTEMS, 35., 2022. **Proceedings** […]. [*S. l.*]: Curran Associates, 2022. DOI: 10.48550/arXiv.2201.11903.

WOLF, Thomas *et al*. Transformers: state-of-the-art natural language processing. *In*: CONFERENCE ON EMPIRICAL METHODS IN NATURAL LANGUAGE PROCESSING: SYSTEM DEMONSTRATIONS, 2020. **Proceedings** […]. [*S. l.*]: Association for Computational Linguistics, 2020. p. 38-45. DOI: 10.18653/v1/2020.emnlp-demos.6.

YANG, An *et al*. Qwen3 technical report. **arXiv**, 2505.09388, 2025. DOI: 10.48550/arXiv.2505.09388.

ZHENG, Lianmin *et al*. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. *In*: ADVANCES IN NEURAL INFORMATION PROCESSING SYSTEMS, 36., 2023. **Proceedings** […]. [*S. l.*]: Curran Associates, 2023. DOI: 10.48550/arXiv.2306.05685.